MIL-M-38510/326B
24 July 1984
SUPERSEDING
MIL-M-38510/326A(USAF)
27 March 1981

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, DEMULTIPLEXERS, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, demultiplexer microcircuits. Two product assurance classes and a choice of case outline and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual 2-line to 4-line decoder/demultiplexer (totem pole output)
02	Dual 2-line to 4-line decoder/demultiplexer (open collector output)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package, option A

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA dc to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation, P_D 1/	55 mW dc
Output current	4 mA dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases E and F	(See MIL-M-38510, Appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

2/ When a thermal resistance value is included in MIL-M-38510, Appendix C, it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC}) - - - - - - - - - - - - - - -	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH}) - - - -	2.0 V dc
Maximum low level input voltage (V_{IL})- - - - -	0.7 V dc
Case operating temperature range (T_C)- - - - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number II (see MIL-M-35810, appendix E).

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TABLE I. Electrical characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 2.0 V, I_{OH} = -400 µA	01	2.5	---	V
Collector cutoff current	I_{CEX}	V_{CC} = 4.5 V, V_{IN} = 2.0 V, V_{OH} = 5.5 V	02	---	100	µA
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V, I_{OL} = 4 mA	All	---	0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = 25°C	All	---	-1.5	V
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	All	-.03	-.38	mA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All	---	20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All	---	100	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IH} = 5.5 V, V_{IL} = GND	01	-6	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	All	---	10	mA
Propagation delay time, low-to-high level A > Y (2 logic levels)	t_{PLH1}	V_{CC} = 5.0 V, R_L = 2.0 kΩ ±10%, C_L = 50 pF ±10%	01	2	26	ns
			02	2	65	
Propagation delay time, low-to-high level B > Y (2 logic levels)	t_{PLH2}		01	2	26	ns
			02	2	65	
Propagation delay time, low-to-high level 1G > 1YO	t_{PLH3}		01	2	26	ns
			02	2	58	
Propagation delay time, low-to-high level 2G > 2YO	t_{PLH4}		01	2	26	ns
			02	2	58	
Propagation delay time, low-to-high level 2C > 2YO	t_{PLH5}		01	2	26	ns
			02	2	58	
Propagation delay time, low-to-high level B > Y (3 logic levels)	t_{PLH6}		01	2	40	ns
			02	2	70	

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TABLE I. Electrical characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time, low-to-high level A → Y (3 logic levels)	t_{PLH7}	V_{CC} = 5.0 V, R_L = 2.0 kΩ ±10%, C_L = 50 pF ±10%	01	2	40	ns
			02	2	70	
Propagation delay time, low-to-high level 1C → 1Y0	t_{PLH8}		01	2	42	ns
			02	2	70	
Propagation delay time, high-to-low level A → Y (2 logic levels)	t_{PHL1}		01	2	46	ns
			02	2	73	
Propagation delay time, high-to-low level B → Y (2 logic levels)	t_{PHL2}		01	2	46	ns
			02	2	73	
Propagation delay time, high-to-low level 1G → 1Y0	t_{PHL3}		01	2	46	ns
			02	2	73	
Propagation delay time, high-to-low level 2G → 2Y0	t_{PHL4}		01	2	46	ns
			02	2	73	
Propagation delay time, high-to-low level 2C → 2Y0	t_{PHL5}		01	2	46	ns
			02	2	73	
Propagation delay time, high-to-low level B → Y (3 logic levels)	t_{PHL6}		01	2	46	ns
			02	2	72	
Propagation delay time, high-to-low level A → Y (3 logic levels)	t_{PHL7}		01	2	46	ns
			02	2	72	
Propagation delay time, high-to-low level 1C → 1Y0	t_{PHL8}		01	2	43	ns
			02	2	69	

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8, 9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	N/A
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D, using the circuit shown on figure 4, or equivalent.

2. T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

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Device types 01, 02

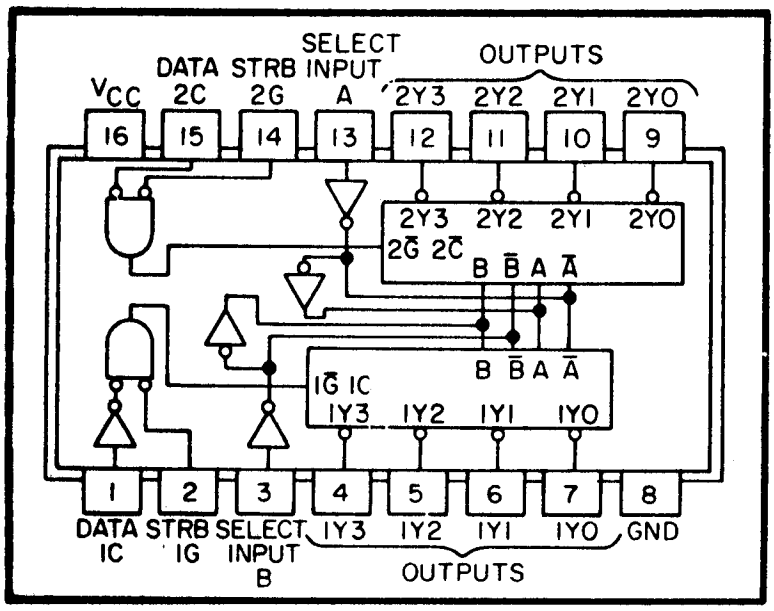


FIGURE 1. Terminal connections.

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Device types 01 and 02

Case 2

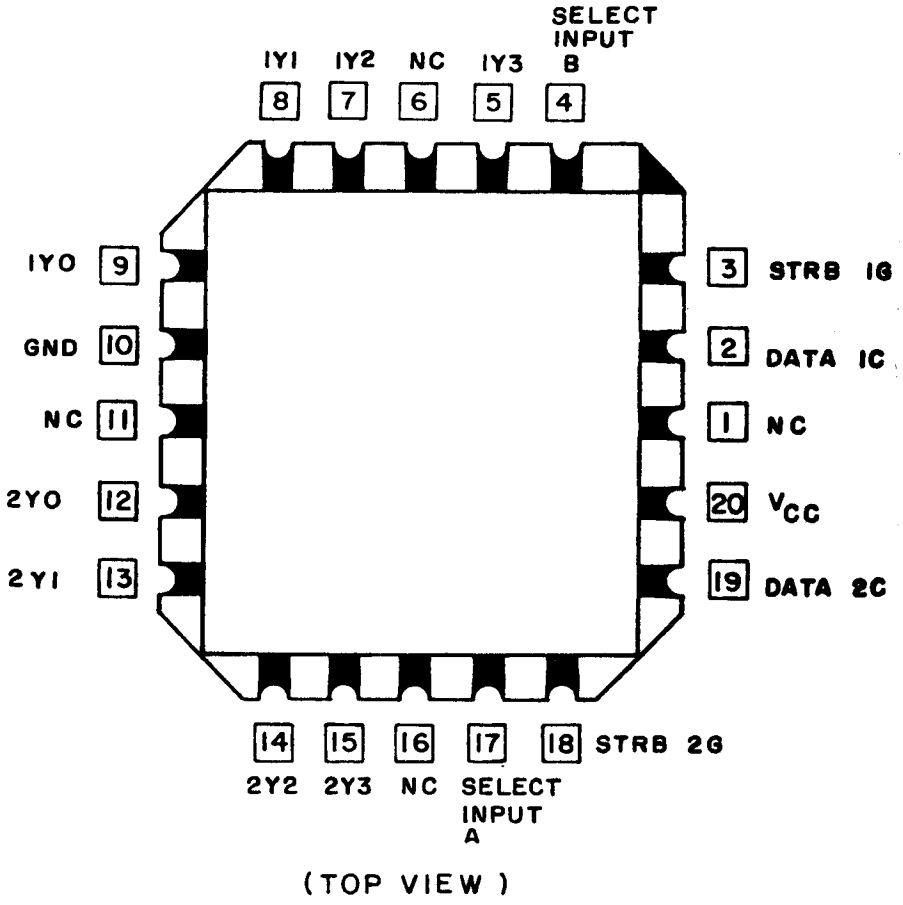


FIGURE 1. Terminal connections. - Continued.

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Device types 01, 02

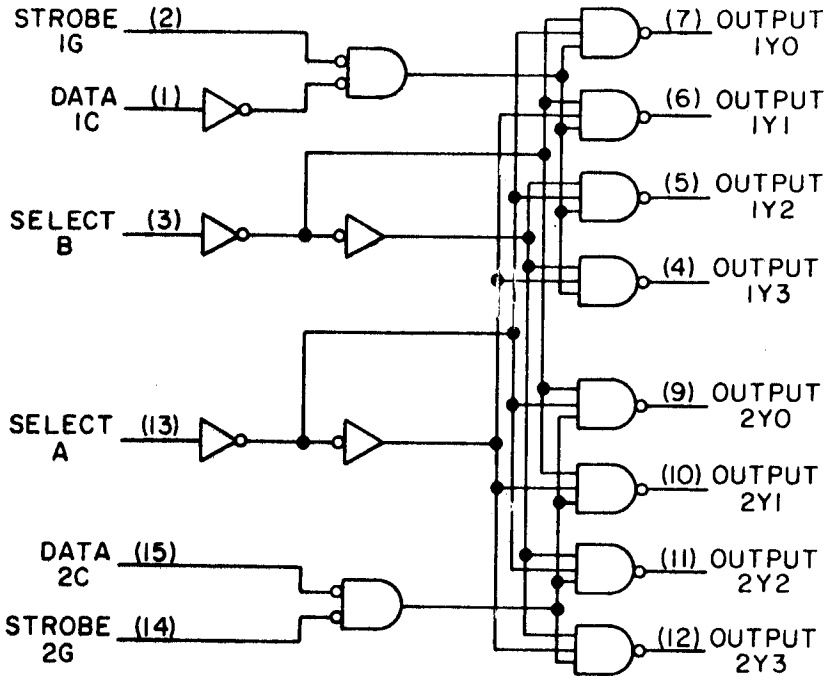


FIGURE 2. Logic diagrams.

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Device types 01, 02

Function Tables
2-line-to-4-line decoder
or 1-line-to-4-line demultiplexer

Inputs				Outputs			
Select		Strobe	Data	1Y0	1Y1	1Y2	1Y3
B	A	1G	1C				
X	X	H	X	H	H	H	H
L	L	L	H	L	H	H	H
L	H	L	H	H	L	H	H
H	L	L	H	H	H	L	H
H	H	L	H	H	H	H	L
X	X	X	L	H	H	H	H

Inputs				Outputs			
Select		Strobe	Data	2Y0	2Y1	2Y2	2Y3
B	A	2G	2C				
X	X	H	X	H	H	H	H
L	L	L	L	L	H	H	H
L	H	L	L	H	L	H	H
H	L	L	L	H	H	L	H
H	H	L	L	H	H	H	L
X	X	X	H	H	H	H	H

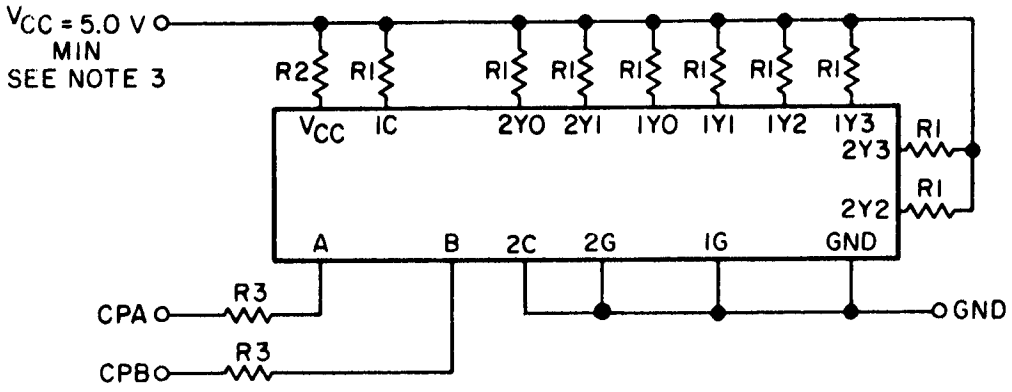
Function Table
3-line-to-8-line decoder
or 1-line-to-8-line demultiplexer

Inputs				Outputs							
Select			Strobe or data	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
C+	B	A	G‡	2Y0	2Y1	2Y2	2Y3	1Y0	1Y1	1Y2	1Y3
X	X	X	H	H	H	H	H	H	H	H	H
L	L	L	L	L	H	H	H	H	H	H	H
L	L	H	L	H	L	H	H	H	H	H	H
L	H	L	L	H	H	L	H	H	H	H	H
L	H	H	L	H	H	H	L	H	H	H	H
H	L	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	L	H	H
H	H	L	L	H	H	H	H	H	H	L	H
H	H	H	L	H	H	H	H	H	H	H	L

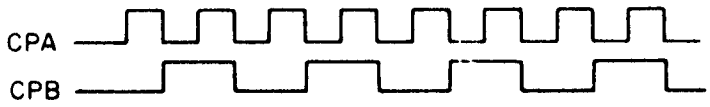
+C = Inputs 1C and 2C connected together.
‡G = Inputs 1G and 2G connected together.
H = High level, L = low level, X = irrelevant.

FIGURE 3. Truth tables.

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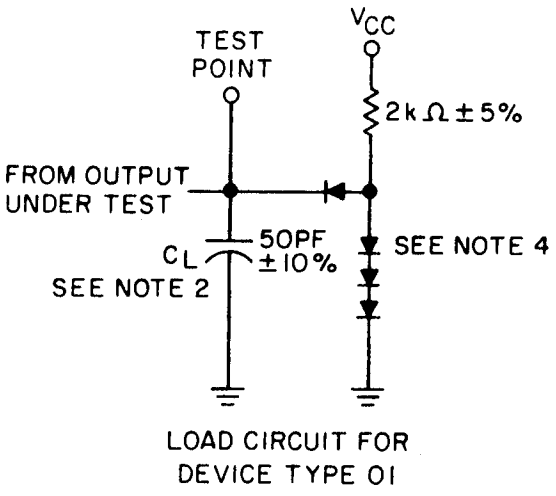
1. CPA = 100 kHz ±50% square wave; duty cycle = 50% ±15%;
   V_{IL} = -0.5 V minimum to 0.7 maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same characteristics as CPA except 1/2 the frequency and phase relationship shown below.



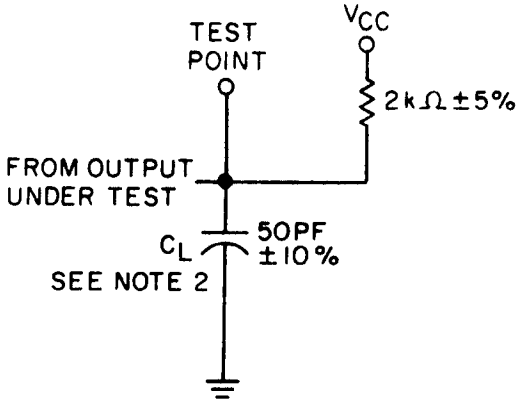
2. RI = 1 kΩ ±5%; R2 = 10Ω ±5%; R3 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure 5.0 V at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuits.

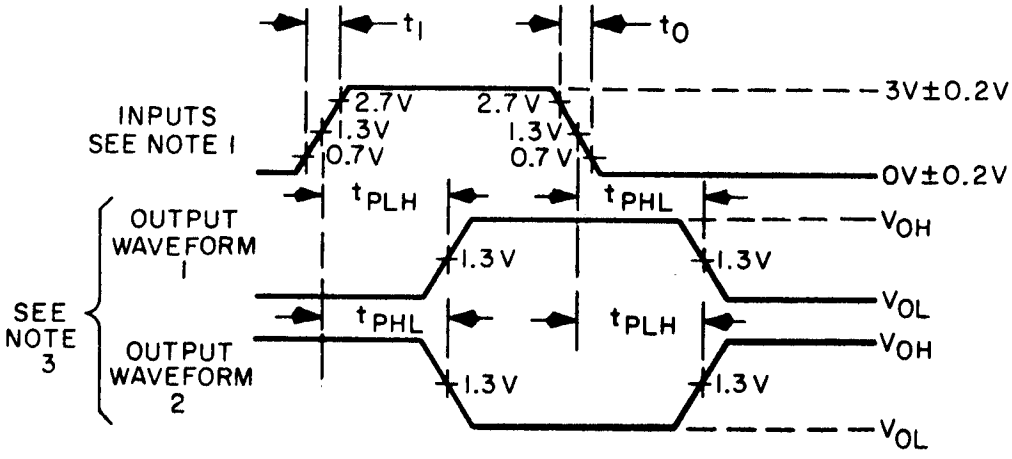
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LOAD CIRCUIT FOR DEVICE TYPE 01



LOAD CIRCUIT FOR DEVICE TYPE 02



NOTES:
1. The pulse generator has the following characteristics: $t_1 \leq 15$ ns to ≤ 6 ns, PRR ≤ 1 MHz.
2. C_L includes probe and jig capacitance.
3. Input - output waveform combination in accordance with the truth tables (see figure 3).
4. All diodes are 1N3064, or equivalent.

FIGURE 5. Switching times for device types 01 and 02.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1		2.0 V					-400 µA	GND								4.5 V	1Y0	2.5		V
		"	2		"				-400 µA		"								"	1Y1	"		"
		"	3		"			-400 µA			"								"	1Y2	"		"
		"	4		"		-400 µA				"								"	1Y3	"		"
		"	5								"	-400 µA							"	2Y0	"		"
		"	6								"		-400 µA				2.0 V		"	2Y1	"		"
		"	7								"			-400 µA			"		"	2Y2	"		"
		"	8								"				-400 µA		"		"	2Y3	"		"
	V_{OL}	3007	9	2.0 V	0.7 V	0.7 V				4 mA	"					0.7 V			"	1Y0		0.4	"
		"	10	"	"	0.7 V			4 mA		"					2.0 V			"	1Y1		"	"
		"	11	"	"	2.0 V		4 mA			"					0.7 V			"	1Y2		"	"
		"	12	"	"	2.0 V	4 mA				"					2.0 V			"	1Y3		"	"
		"	13			0.7 V					"	4 mA				0.7 V	0.7 V	0.7 V	"	2Y0		"	"
		"	14			0.7 V					"		4 mA			2.0 V	"	"	"	2Y1		"	"
		"	15			2.0 V					"			4 mA		0.7 V	"	"	"	2Y2		"	"
		"	16			2.0 V					"				4 mA	2.0 V	"	"	"	2Y3		"	"
	V_{IC}		17	-18 mA							"								"	1C		-1.5	"
			18		-18 mA						"								"	1G		"	"
			19			-18 mA					"								"	B		"	"
			20								"					-18 mA			"	A		"	"
			21								"						-18 mA		"	2G		"	"
			22								"							-18 mA	"	2C		"	"
	I_{IL1}	3009	23	0.4 V							"								5.5 V	1C	2/	2/	mA
			24		0.4 V						"								"	1G	"	"	"
			25			0.4 V					"								"	B	"	"	"
			26								"					0.4 V			"	A	"	"	"
			27								"						0.4 V		"	2G	"	"	"
			28								"							0.4 V	"	2C	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	1C		20	µA
			30		2.7 V						"								"	1G		"	"
			31			2.7 V					"								"	B		"	"
			32								"					2.7 V			"	A		"	"
			33								"						2.7 V		"	2G		"	"
			34								"							2.7 V	"	2C		"	"
	I_{IH2}		35	5.5 V							"								"	1C		100	"
			36		5.5 V						"								"	1G		"	"
			37			5.5 V					"								"	B		"	"
			38								"					5.5 V			"	A		"	"
			39								"						5.5 V		"	2G		"	"
			40								"							5.5 V	"	2C		"	"
	I_{OS}	3011	41		5.5 V					GND	"								"	1Y0	3/	3/	mA
			42		5.5 V				GND		"								"	1Y1	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OS}	3011	43		5.5 V		GND	GND			GND								5.5 V	1Y2	3/	3/	mA
		"	44								"								"	1Y3	"	"	"
		"	45								"	GND							"	2Y0	"	"	"
		"	46								"		GND					5.5 V	"	2Y1	"	"	"
		"	47								"			GND				"	"	2Y2	"	"	"
		"	48								"				GND			"	"	2Y3	"	"	"
	I_{CC}	3005	49	4.5 V	GND	4.5 V					"					4.5 V	GND	GND	"	V_{CC}		10.0	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C 4/	Truth table tests	3014	50	B	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	5.0 V		5/		
		"	51	"	"	B	"	"	"	"	"	"	"	"	"	A	"	"	"		"		
		"	52	"	"	A	"	"	"	"	"	"	"	"	"	B	"	"	"		"		
		"	53	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"		"		
		"	54	A	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"		"		
		"	55	"	B	"	"	"	"	L	"	L	"	"	"	"	B	"	"		"		
		"	56	"	"	"	"	"	L	H	"	H	L	"	"	A	"	"	"		"		
		"	57	"	"	A	"	L	H	"	"	"	H	L	"	B	"	"	"		"		
		"	58	"	"	"	L	H	"	"	"	"	"	H	L	A	"	"	"		"		
		"	59	B	"	"	H	"	"	"	"	"	"	"	H	A	"	A	"		"		
		"	60	B	A	"	H	"	"	"	"	"	"	"	H	A	A	A	"		"		
		"	61	B	A	B	H	"	"	"	"	"	"	"	H	B	A	A	"		"		
8	Same tests and terminal conditions as for subgroup 7, except T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	62	4.5 V	GND	GND				OUT	GND					IN			5.0 V	A > 1Y0	2	20	ns
		"	63			GND					"	OUT				"	GND	GND	"	A > 2Y0	"	"	"
		"	64	4.5 V	GND	4.5 V		OUT			"					"			"	A > 1Y2	"	"	"
		"	65			4.5 V					"			OUT		"	GND	GND	"	A > 2Y2	"	"	"
	t_{PLH2}	"	66	4.5 V	GND	IN				OUT	"					GND			"	B > 1Y0	"	"	"
		"	67			"					"	OUT				GND	GND	GND	"	B > 2Y0	"	"	"
		"	68	4.5 V	GND	"			OUT		"					4.5 V			"	B > 1Y1	"	"	"
		"	69			"					"		OUT			4.5 V	GND	GND	"	B > 2Y1	"	"	"
	t_{PLH3}	"	70	4.5 V	IN	GND				OUT	"					GND			"	1G > 1Y0	"	"	"
	t_{PLH4}	"	71			GND					"	OUT				GND	IN	GND	"	2G > 2Y0	"	"	"
	t_{PLH5}	"	72			GND					"	OUT				GND	GND	IN	"	2C > 2Y0	"	"	"
	t_{PLH6}	"	73	4.5 V	GND	IN		OUT			"					GND			"	B > 1Y2	"	31	"
		"	74			"					"			OUT		GND	GND	GND	"	B > 2Y2	"	"	"
		"	75	4.5 V	GND	"	OUT				"					4.5 V			"	B > 1Y3	"	"	"
		"	76			"					"				OUT	4.5 V	GND	GND	"	B > 2Y3	"	"	"
	t_{PLH7}	"	77	4.5 V	GND	GND			OUT		"					IN			"	A > 1Y1	"	"	"
		"	78			GND					"		OUT			"	GND	GND	"	A > 2Y1	"	"	"
		"	79			4.5 V					"				OUT	"	GND	GND	"	A > 2Y3	"	"	"
		"	80	4.5 V	GND	4.5 V	OUT				"					"			"	A > 1Y3	"	"	"
	t_{PLH8}	"	81	IN	GND	GND				OUT	"					GND			"	1C > 1Y0	"	32	"
	t_{PHL1}	"	82	4.5 V	GND	GND				OUT	"					IN			"	A > 1Y0	"	35	"
		"	83			GND					"	OUT				"	GND	GND	"	A > 2Y0	"	"	"
		"	84	4.5 V	GND	4.5 V		OUT			"					"			"	A > 1Y2	"	"	"
		"	85			4.5 V					"			OUT		"	GND	GND	"	A > 2Y2	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test Limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	1C	1G	5	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PHL2}	3003 Fig. 5	86	4.5 V	GND	IN				OUT	GND					GND			5.0 V	B > 1Y0	2	35	ns
		"	87			"					"	OUT				GND	GND	GND	"	B > 2Y0	"	"	"
		"	88	4.5 V	GND	"			OUT		"					4.5 V			"	B > 1Y1	"	"	"
		"	89			"					"		OUT			4.5 V	GND	GND	"	B > 2Y1	"	"	"
	t_{PHL3}	"	90	4.5 V	IN	GND				OUT	"					GND			"	1G > 1Y0	"	"	"
	t_{PHL4}	"	91			GND					"	OUT				"	IN	GND	"	2G > 2Y0	"	"	"
	t_{PHL5}	"	92			GND					"	OUT				"	GND	IN	"	2C > 2Y0	"	"	"
	t_{PHL6}	"	93	4.5 V	GND	IN		OUT			"					"			"	B > 1Y2	"	"	"
		"	94			"					"			OUT		"	GND	GND	"	B > 2Y2	"	"	"
		"	95	4.5 V	GND	"	OUT				"					4.5 V			"	B > 1Y3	"	"	"
		"	96			"					"				OUT	4.5 V	GND	GND	"	B > 2Y3	"	"	"
	t_{PHL7}	"	97	4.5 V	GND	GND			OUT		"					IN			"	A > 1Y1	"	"	"
		"	98			GND					"		OUT			"	GND	GND	"	A > 2Y1	"	"	"
		"	99			4.5 V					"				OUT	"	GND	GND	"	A > 2Y3	"	"	"
		"	100	4.5 V	GND	4.5 V	OUT				"					"			"	A > 1Y3	"	"	"
	t_{PHL8}	"	101	IN	GND	GND				OUT	"					GND			"	1C > 1Y0	"	33	"
10 T_C = +125°C	t_{PLH1}	"	102-105																		"	26	"
	t_{PLH2}	"	106-109																		"	"	"
	t_{PLH3}	"	110																		"	"	"
	t_{PLH4}	"	111																		"	"	"
	t_{PLH5}	"	112																		"	"	"
	t_{PLH6}	"	113-116																		"	40	"
	t_{PLH7}	"	117-120																		"	40	"
	t_{PLH8}	"	121	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																	"	42	"
	t_{PHL1}	"	122-125																		"	46	"
	t_{PHL2}	"	126-129																		"	46	"
	t_{PHL3}	"	130																		"	"	"
	t_{PHL4}	"	131																		"	"	"
	t_{PHL5}	"	132																		"	"	"
	t_{PHL6}	"	133-136																		"	"	"
	t_{PHL7}	"	137-140																		"	"	"
	t_{PHL8}	"	141																		"	43	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case 2 pins not referenced are NC.

2/

Test	Min/Max limits in (mA) for circuits					
	A	B	C	D	E	F
I_{IL1}	-.12/ -.36	-.03/ -.30	-.15/ -.38		-.10/ -.34	-.12/ -.36

3/

Test	Min/Max limits in (mA) for circuits					
	A	B	C	D	E	F
I_{OS}	-6/ -10	-15/ -100	-20/ -100		-15/ -100	-15/ -100

4/ Inputs: A = 3.0 V min; B = 0.0 V or GND.
5/ Outputs: H > 1.5 V; L < 1.5 V

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TABLE III. Group A i ction for device type 02. 1/
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 2/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
1 T_C = +25°C	I_{CEX}		1		2.0 V						GND								4.5 V	1Y0		100	µA
			2		"						"								"	1Y1		"	"
			3		"			5.5 V			"								"	1Y2		"	"
			4				5.5 V				"								"	1Y3		"	"
			5								"	5.5 V							"	2Y0		"	"
			6						5.5 V		"		5.5 V				2.0 V		"	2Y1		"	"
			7								"			5.5 V			"		"	2Y2		"	"
			8								"				5.5 V		"		"	2Y3		"	"
	V_{OL}	3007	9	2.0 V	0.7 V	0.7 V				4 mA	"					0.7 V			"	1Y0		0.4	V
		"	10	"	"	0.7 V				"	"					2.0 V			"	1Y1		"	"
		"	11	"	"	2.0 V		4 mA	4 mA	"	"					0.7 V			"	1Y2		"	"
		"	12			2.0 V	4 mA			"	"	4 mA				2.0 V			"	1Y3		"	"
		"	13			0.7 V				"	"		4 mA			0.7 V	0.7 V	0.7 V	"	2Y0		"	"
		"	14			0.7 V				"	"			4 mA		2.0 V	"	"	"	2Y1		"	"
		"	15			2.0 V				"	"				4 mA	0.7 V	"	"	"	2Y2		"	"
		"	16			2.0 V				"	"					2.0 V			"	2Y3		"	"
	V_{IC}		17	-18 mA							"								"	1C		-1.5	"
			18		-18 mA						"								"	1G		"	"
			19			-18 mA					"								"	B		"	"
			20								"					-18 mA			"	A		"	"
			21								"						-18 mA		"	2G		"	"
			22								"							-18 mA	"	2C		"	"
	I_{IL1}	3009	23	0.4 V							"								5.5 V	1C	3/	3/	mA
		"	24		0.4 V						"								"	1G	"	"	"
		"	25			0.4 V					"								"	B	"	"	"
		"	26								"					0.4 V			"	A	"	"	"
		"	27								"						0.4 V		"	2G	"	"	"
		"	28								"							0.4 V	"	2C	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	1C		20	µA
		"	30		2.7 V						"								"	1G		"	"
		"	31			2.7 V					"								"	B		"	"
		"	32								"					2.7 V			"	A		"	"
		"	33								"						2.7 V		"	2G		"	"
		"	34								"							2.7 V	"	2C		"	"
	I_{IH2}	"	35	5.5 V							"								"	1C		100	"
		"	36		5.5 V						"								"	1G		"	"
		"	37			5.5 V					"								"	B		"	"
		"	38								"					5.5 V			"	A		"	"
		"	39								"						5.5 V		"	2G		"	"
		"	40								"							5.5 V	"	2C		"	"
	I_{CC}	3005	41	4.5 V	GND	4.5 V					"					4.5 V	GND	GND	"	V_{CC}		10.0	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued. 1/
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 2/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}		Min	Max	
7 T_C = +25°C 4/	Truth table tests	3014	42	B	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	5.0 V		5/		
		"	43	"	"	B	"	"	"	"	"	"	"	"	"	A	"	"	"		"		
		"	44	"	"	A	"	"	"	"	"	"	"	"	"	B	"	"	"		"		
		"	45	"	"	B	"	"	"	"	"	"	"	"	"	"	"	B	"		"		
		"	46	A	"	"	"	"	"	L	"	L	"	"	"	"	B	"	"		"		
		"	47	"	B	"	"	"	L	H	"	H	L	"	"	A	"	"	"		"		
		"	48	"	"	A	"	L	H	"	"	"	L	H	L	B	"	"	"		"		
		"	49	"	"	"	L	H	"	"	"	"	H	"	H	A	"	A	"		"		
		"	50	"	"	"	H	"	"	"	"	"	"	H	L	A	A	"	"		"		
		"	51	B	"	"	H	"	"	"	"	"	"	"	H	A	A	"	"		"		
		"	52	B	A	"	H	"	"	"	"	"	"	"	H	A	A	"	"		"		
		"	53	B	A	B	H	"	"	"	"	"	"	"	H	B	A	A	"		"		
8	Same tests and terminal conditions as for subgroup 7, except T_C = 125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	54	4.5 V	GND	GND				OUT	GND					IN			5.0 V	A → 1Y0	2	50	ns
		"	55			GND					"					"	GND	GND	"	A → 2Y0	"	"	"
		"	56	4.5 V	GND	4.5 V		OUT			"	OUT				"			"	A → 1Y2	"	"	"
		"	57			4.5 V					"			OUT		"	GND	GND	"	A → 2Y2	"	"	"
	t_{PLH2}	"	58	4.5 V	GND	IN				OUT	"					GND			"	B → 1Y0	"	"	"
		"	59			"					"	OUT				GND	GND	GND	"	B → 2Y0	"	"	"
		"	60	4.5 V	GND	"			OUT		"					4.5 V			"	B → 1Y1	"	"	"
		"	61			"					"		OUT			4.5 V	GND	GND	"	B → 2Y1	"	"	"
	t_{PLH3}	"	62	4.5 V	IN	GND				OUT	"					GND			"	1G → 1Y0	"	45	"
	t_{PLH4}	"	63			GND					"	OUT				GND	IN	GND	"	2G → 2Y0	"	"	"
	t_{PLH5}	"	64			GND					"	OUT				GND	GND	IN	"	2C → 2Y0	"	"	"
	t_{PLH6}	"	65	4.5 V	GND	IN		OUT			"					GND			"	B → 1Y2	"	54	"
		"	66			"					"			OUT		GND	GND	GND	"	B → 2Y2	"	"	"
		"	67	4.5 V	GND	"	OUT				"					4.5 V			"	B → 1Y3	"	"	"
		"	68			"					"				OUT	4.5 V	GND	GND	"	B → 2Y3	"	"	"
	t_{PLH7}	"	69	4.5 V	GND	GND			OUT		"					IN			"	A → 1Y1	"	"	"
		"	70			GND					"		OUT			"	GND	GND	"	A → 2Y1	"	"	"
		"	71			4.5 V					"				OUT	"	GND	GND	"	A → 2Y3	"	"	"
		"	72	4.5 V	GND	4.5 V	OUT				"					"			"	A → 1Y3	"	"	"
	t_{PLH8}	"	73	IN	GND	GND				OUT	"					GND			"	1C → 1Y0	"	"	"
	t_{PHL1}	"	74	4.5 V	GND	GND				OUT	"					IN			"	A → 1Y0	"	56	"
		"	75			GND					"	OUT				"	GND	GND	"	A → 2Y0	"	"	"
		"	76	4.5 V	GND	4.5 V		OUT			"					"			"	A → 1Y2	"	"	"
		"	77			4.5 V					"			OUT		"	GND	GND	"	A → 2Y2	"	"	"
	t_{PHL2}	"	78	4.5 V	GND	IN				OUT	"					GND			"	B → 1Y0	"	"	"
		"	79			"					"	OUT				GND	GND	GND	"	B → 2Y0	"	"	"
		"	80	4.5 V	GND	"			OUT		"					4.5 V			"	B → 1Y1	"	"	"
		"	81			"					"		OUT			4.5 V	GND	GND	"	B → 2Y1	"	"	"
	t_{PHL3}	"	82	4.5 V	IN	GND				OUT	"					GND			"	1G → 1Y0	"	"	"
	t_{PHL4}	"	83			GND					"	OUT				GND	IN	GND	"	2G → 2Y0	"	"	"
	t_{PHL5}	"	84			GND					"	OUT				GND	GND	IN	"	2C → 2Y0	"	"	"
	t_{PHL6}	"	85	4.5 V	GND	IN		OUT			"					GND			"	B → 1Y2	"	"	"
		"	86			"					"			OUT		GND	GND	GND	"	B → 2Y2	"	"	"
		"	87	4.5 V	GND	"	OUT				"					4.5 V			"	B → 1Y3	"	"	"
		"	88			"					"				OUT	4.5 V	GND	GND	"	B → 2Y3	"	"	"
	t_{PHL7}	"	89	4.5 V	GND	GND			OUT		"					IN			"	A → 1Y1	"	"	"
		"	90			GND					"		OUT			"	GND	GND	"	A → 2Y1	"	"	"
		"	91			4.5 V					"				OUT	"	GND	GND	"	A → 2Y3	"	"	"
		"	92	4.5 V	GND	4.5 V	OUT				"					"			"	A → 1Y3	"	"	"
	t_{PHL8}	"	93	IN	GND	GND				OUT	"					GND			"	1C → 1Y0	"	53	"

See footnotes at end of device type 02.

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TABLE III. Group A ins on for device type 02 - Continued. 1/
Terminal conditions (pins not desi...ted may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 2/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}		Min	Max	
10 T_C =+125°C	t_{PLH1}	3003 Fig. 5	94-97																		2	65	ns
	t_{PLH2}	"	98-101																		"	65	"
	t_{PLH3}	"	102																		"	58	"
	t_{PLH4}	"	103																		"	"	"
	t_{PLH5}	"	104																		"	"	"
	t_{PLH6}	"	105-108																		"	70	"
	t_{PLH7}	"	109-112																		"	70	"
	t_{PLH8}	"	113																		"	70	"
	t_{PHL1}	"	114-117																		"	73	"
	t_{PHL2}	"	118-121																		"	73	"
	t_{PHL3}	"	122																		"	"	"
	t_{PHL4}	"	123																		"	"	"
	t_{PHL5}	"	124																		"	"	"
	t_{PHL6}	"	125-128																		"	72	"
	t_{PHL7}	"	129-132																		"	72	"
	t_{PHL8}	"	133																		"	69	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.

1/ A pullup resistor of 0.5 kΩ to 2 kΩ shall be placed from each output to V_{CC} (pin 16).
2/ Case 2 pins not referenced are NC.
3/

Test	Min/Max limits in (mA) for circuits					
	A	B	C	D	E	F
I_{IL1}	-.12/ -.36	-.03/ -.30	-.15/ -.38		-.10/ -.34	-.12/ -.36

4/ Inputs: A = 3.0 V min; B = 0.0 V or GND.
5/ Outputs: H > 1.5 V; L < 1.5 V

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

1. Test condition D, using the circuit shown on figure 4, or equivalent.

2. T_A = +125°C minimum.

3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510, and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential
I_{IN}	Current flowing into an input terminal
V_{IN}	Voltage level at an input terminal

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6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS155
02	54LS156

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.7 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

Devices	Texas Instruments	Signetics	National	Raytheon	Fairchild	Motorola
	A	B	C	D	E	F
01	X	X	X		X	X
02	X	X	X		X	X

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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